WORKED SOLUTION

STEP I 1987, Question 3

Pure • Separable Differential Equations

STEP I 1987, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s1-q3

The key idea

The equation looks unpleasant in v, but the left-hand side is begging to be recognised: $x^3v' + x^2v$ is x^2 times the derivative of xv. So the suggested substitution turns the left-hand side into x^2y', and the right-hand side, rewritten in terms of y, separates neatly. After that it is partial fractions and a careful choice of sign.

Solution

Changing variable

With $y = xv$ the product rule gives $y' = v + xv'$, so

$$x^3\frac{\mathrm{d}v}{\mathrm{d}x} + x^2v = x^2\left(x\frac{\mathrm{d}v}{\mathrm{d}x} + v\right) = x^2\frac{\mathrm{d}y}{\mathrm{d}x}.$$

On the right, $x^2v^2 = y^2$ and $v = y/x$, so the right-hand side is

$$\frac{1+y^2}{(1+x^2)\,y/x} = \frac{x(1+y^2)}{(1+x^2)\,y}.$$

The equation becomes

$$x^2\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x(1+y^2)}{(1+x^2)\,y}, \qquad \text{that is,} \qquad \frac{y}{1+y^2}\,\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{x(1+x^2)},$$

dividing by x^3 (we work near $x = 1$, so $x > 0$) and multiplying by $y/(1+y^2)$. The variables have separated.

Integrating

The left-hand side integrates to $\frac{1}{2}\ln(1+y^2)$. For the right, partial fractions give

$$\frac{1}{x(1+x^2)} = \frac{1}{x} - \frac{x}{1+x^2},$$

which is easily checked by putting the right-hand side over a common denominator. Hence, for $x > 0$,

$$\tfrac{1}{2}\ln(1+y^2) = \ln x - \tfrac{1}{2}\ln(1+x^2) + C, \qquad \text{so} \qquad 1+y^2 = \frac{Ax^2}{1+x^2}$$

for a positive constant $A = \mathrm{e}^{2C}$.

When $x = 1$ we have $v = 1$, so $y = 1$ too, giving $2 = A/2$ and $A = 4$. Therefore

$$y^2 = \frac{4x^2}{1+x^2} - 1 = \frac{3x^2-1}{1+x^2}, \qquad v^2 = \frac{y^2}{x^2} = \frac{3x^2-1}{x^2(1+x^2)}.$$

? **Ask yourself** Which square root do we take, and where is the solution valid?

The original equation has v in a denominator, so a solution can never pass through $v = 0$; being continuous, it keeps the sign it has at $x = 1$, which is positive. So we take the positive root:

$$\boxed{v(x) = \frac{\sqrt{3x^2-1}}{x\sqrt{1+x^2}}}$$

valid for $x > 1/\sqrt{3}$. As x decreases to $1/\sqrt{3}$, v falls to 0 and the equation breaks down, so the solution cannot be continued below that point.

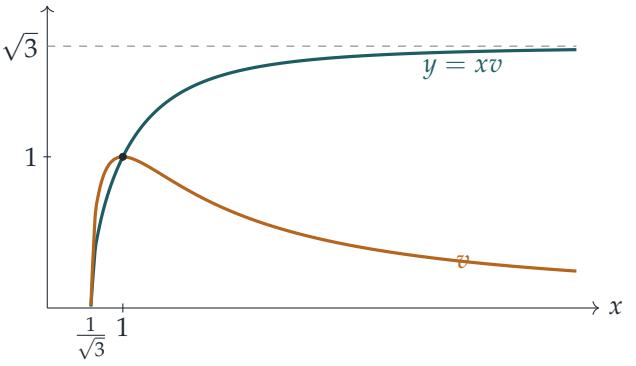
Large x

Dividing the top and bottom by x^2,

$$v(x) = \frac{1}{x}\cdot\frac{\sqrt{3-1/x^2}}{\sqrt{1+1/x^2}} \approx \frac{\sqrt{3}}{x},$$

so $v(x) \to 0$ as $x \to \infty$. Equivalently, $y = xv \to \sqrt{3}$: the graph of y has the horizontal asymptote $y = \sqrt{3}$, and $v = y/x$ decays like $\sqrt{3}/x$.

In fact the starting point is the peak: writing $u = x^2$, the derivative of $(3u-1)/(u^2+u)$ has numerator $-(3u+1)(u-1)$, so v rises to its maximum value 1 at $x = 1$ and then decreases towards 0.



Checks

- At $x = 1$: $v = \sqrt{2}/(1 \cdot \sqrt{2}) = 1$, as required.
- Substituting back: from $1 + y^2 = 4x^2/(1 + x^2)$, implicit differentiation gives $2yy' = 8x/(1 + x^2)^2$, and $(1 + y^2)/(x(1 + x^2)) = 4x/(1 + x^2)^2$, so $yy'/(1 + y^2) = 1/(x(1 + x^2))$ as it should.
- Numerically, stepping the original equation from $(1, 1)$ with a Runge–Kutta method gives $v(10) \approx 0.172058$, matching the formula, and at $x = 10^4$ the formula gives $xv \approx 1.73205 \approx \sqrt{3}$.

Common slips

- Answering $\sqrt{3}$ for the limit: that is the limit of y, but the question asks about v, which tends to 0.
- Exponentiating the constant wrongly: $\ln(1 + y^2) = \ln \frac{x^2}{1+x^2} + 2C$ gives $1 + y^2 = Ax^2/(1 + x^2)$, a multiplicative constant, not $x^2/(1 + x^2) + 2C$.
- Taking $\pm$ in the square root without deciding: the condition $v(1) = 1$ and the ban on $v = 0$ fix the positive sign.

Beyond STEP

The substitution here works because the left-hand side is an exact derivative after multiplying through, the same idea that lies behind integrating factors and homogeneous equations. The care needed over the sign and the domain is a first glimpse of existence and uniqueness theory: a solution through a given point exists until it runs into a place where the equation stops making sense, here $v = 0$. Boyce and DiPrima, *Elementary Differential Equations*, treats separable equations and the question of how far a solution extends.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.